CP115
Computer Programming 1
Semester 1
2026/2027
2 hours 30 minutes

CP115
Pengaturcaraan Komputer 1
Semester 1
Sesi 2026/2027
2 jam 30 minit

Nama :
Tutoran :

KOLEJ MATRIKULASI PERAK
PERAK MATRICULATION COLLEGE

PRA-PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME TRIAL EXAMINATION

PENGATURCARAAN KOMPUTER
2 jam 30 minit

DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.
JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.

Untuk Kegunaan Pemeriksa *For Examiner's Use*		
No. Soalan *Question No.*	Markah *Mark*	
	Markah Penuh *Full Mark*	Pemeriksa *Examiner*
1	25	
2	10	
3	13	
4	15	
5	15	
6	12	
JUMLAH	90	

1 (a) Problem-solving in programming involves several important steps. Identify the correct problem-solving step based on the descriptions below.

[2 *marks*]

	Description	**Problem solving step**
(i)	A detailed explanation describing how a completed food-delivery application works and how users interact with its features.	
(ii)	Running the application with different test cases to ensure that the program produces the expected results and identifying any errors.	

(b) Determine the type of errors based on the given scenarios.

[3 *mark*s]

	Scenario	**Type of errors**
(i)	Azfarzarul writes if followers > 1000 but forgets to put : at the end of the statement.	
(ii)	Johnny wants to display "Viral" when the number of views is 10,000 or more, but accidentally writes views < 10000.	
(iii)	A program attempts to open concert_ticket.csv, but the file does not exist in the specified folder.	

(c) Define the following components of the IPO model based on the given scenario.

[3 *mark*s]

	Scenario	**IPO Definition**
(i)	The system determines the delivery fee based on the distance, applies a discount for eligible members, and calculates the final amount to be paid	

(ii)	The final payment for each order	
(iii)	The customer enters the food price, delivery distance, and membership status.	

(d) Determine whether each scenario represents Input, Process or Output.

[5 *mark*s]

	Scenario	**Input, Process or Output**
(i)	A user enters a discount code in an online shopping application.	
(ii)	The system calculates the final price after applying the discount code.	
(iii)	The application displays "RM45.90 – Order Confirmed".	
(iv)	A user selects the number of concert tickets to purchase.	
(v)	The system displays the QR code for entry into the concert.	

(e) Identify the input, process and output.

(i) A food-delivery application asks a customer to enter the number of meals ordered. Each meal costs RM18.00. A delivery charge of RM5.00 is added, followed by a 10% service charge calculated based on the meal subtotal. The program calculates and displays the final amount to be paid.

[5 *mark*s]

Input:	
Process:	
Output:	

(ii) A social media application uses a counter-controlled loop to prompt a user for the duration of 5 videos watched during a study break. After collecting all 5 video durations, the program calculates and displays the total viewing time, the average video length and the shortest video watched.

[7 *marks*]

Input:	
Process:	
Output:	

2 (a) A programmer needs to plan the solution to a problem before writing the actual program code. The programmer may use an algorithm, pseudocode and flowchart to represent the proposed solution. Briefly define how it is used to represent a program solution.

[3 *marks*]

Term	**Definition of usage**
Algorithm	
Pseudocode	
Flowchart	

(b) The following pseudocode is used by a fitness application to classify a user's daily step count. The program repeatedly asks the user to enter the number of steps. The program stops when a negative value is entered. Convert the pseudocode into a suitable flowchart.

[7 *marks*]

```
Start
    Read steps
    While steps ≥ 0
        If steps ≥ 10000
            Print "Excellent!"
        Else If steps ≥ 6000
            Print "Good progress!"
        Else
            Print "Try to move more!"
    Endif
    Read steps
    End While
Stop
```

3 (a) Identify the appropriate programming paradigm based on each statement.

[4 *marks*]

	Statement	Programming Paradigm
(i)	A program is developed by breaking a large problem into smaller functions.	
(ii)	The programmer focuses on objects containing data and methods.	
(iii)	The program follows a step-by-step sequence of instructions.	
(iv)	Examples include Java and C++.	

(b) In Python, reserved words have predefined meanings and cannot be used as identifiers such as variable or function names. State any **one (1)** Python reserved words and briefly state the purpose of the word.

[2 *marks*]

(c) State the data type and state the example value for each variable.

[2 *marks*]

Variable name	Value	Data type
is_verified		Boolean
username	kmpk	

(d) The following codes contain errors. Correct the code so that it will display RM1000 properly.

[1 *mark*]

```
Price = 2000
Print (RM, price)
```

(e) A TikiTok creator wants to calculate the total cost of creating a quick story post. The application asks the user to enter the duration of the video in seconds and the cost per second of their editing software. The program then multiplies the video duration by the cost per second to calculate the total editing cost, and finally displays the total cost to the user. Write the python program.

[4 *marks*]

4 (a) Identify the type of control structure represented by each scenario.

[2 *marks*]

Scenario	Type of control structure
A music application checks whether a song is marked as favourite and displays a different message depending on the result.	
A program converts the number of TikiTok minutes into hours and minutes.	

(b) Determine the output of each expression. Assume x = 8 and y = 12.

[6 *marks*]

	Expression	Output
(i)	`x == y`	
(ii)	`x < y - 3`	
(iii)	`x >= y - 4`	
(iv)	`x == y or x > 5`	
(v)	`x != 8 and y > 10`	
(vi)	`y > 0 and y < 20`	

(c) A Kayempikay technology camp uses a screening program to determine a student's status for an Advanced Python Workshop. The student enters the number of programming exercises completed and the percentage score of a programming assessment. The program determines the student's status based on the following conditions:

- Highly Qualified – completed at least 15 exercises and achieved a score of 85 or above.
- Qualified – completed at least 10 exercises and achieved a score of 70 or above.
- Partially Qualified – completed at least 10 exercises but achieved a score below 70.
- Not Qualified – completed fewer than 10 exercises.

Write a Python program to determine and display the appropriate status.

[7 *marks*]

5 (a) A counter-controlled loop contains three important components. Consider the following code:

```
for day in range(1, 8):
    print(day)
```

Rewrite the given coding based on the components update:

Initialize Number to 333

Condition: Number > 3

Decrease the Number by 3

[3 *marks*]

(b) Convert the following for loop into an equivalent while loop.

```
total = 0
i = 0
for count in range(2, 21, 2):
    total += count
    i = i + 1
    print(i)
print(total)
```

[5 *marks*]

(c) A popular study cafe, ZuEmi wants to analyse its daily study-session bookings. The program repeatedly asks the user to enter the number of hours booked for each study session. The program stops when the user enters -1. After all bookings have been entered, the program must display the total number of study sessions, the total number of hours booked, and the average number of hours per session.

```
Sample interaction:
Enter study hours (-1 to finish): 2
Enter study hours (-1 to finish): 4
Enter study hours (-1 to finish): 3
Enter study hours (-1 to finish): -1

Total study sessions: 3
Total study hours: 9
Average study hours: 3.0
```

Write a Python program that meets the requirements.

[7 *marks*]

6 The University of Gopenghegen is organising a Campus Night Festival for students. The organiser wants to develop a Python program to calculate the total ticket cost for a group of friends. The program first asks the user to enter the number of people in the group. For each person, the program asks for their age and determines the ticket price based on the following table.

Age Group	Age Range	Ticket Price
Teen	13–17	RM12
Student	18–25	RM20
Adult	26–59	RM30
Senior	60 and above	RM15

The program calculates the total ticket price for all members. If the total ticket price is more than RM100, a 10% discount is applied. The program should finally display the total number of people, the eldest student age, original ticket price, discount amount and final ticket price.

```
Sample Output
Enter number of people: 5

Enter age: 18
Enter age: 20
Enter age: 21
Enter age: 17
Enter age: 30

--- CAMPUS NIGHT FESTIVAL ---
Total people: 5
The eldest student age: 30
Original ticket price: RM 102.0
Discount: RM 10.2
Final ticket price: RM 91.8
```

Write a Python program that fulfils all the requirements.

[12 *marks*]

END OF QUESTION PAPER